

**INTERNATIONAL CYBERSECURITY
AND DIGITAL FORENSICS ACADEMY**

**INTRO TO CRITICAL INFRASTRUCTURE
FOR IT LEARNERS**

LAB 01 — INTRODUCTION TO PROGRAMMABLE LOGIC CONTROLLERS

Student Edition

Authorized ICDFA training environments only
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Lab 01: Introduction to Programmable Logic Controllers

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 2–3 hours

Submission deadline: As published in ICDFA E-Campus.

Authorization & Safety: Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

Learning Outcomes

- Explain the role of PLCs in industrial and critical-infrastructure environments.
- Identify core PLC components including processor, power supply, input modules, output modules, rack/chassis and expansion modules.
- Differentiate discrete (digital) and analog I/O.
- Launch OpenPLC Editor and inspect a ladder-logic motor-control example.
- Explain the purpose of normally open contacts, normally closed contacts and output coils.

Laboratory Environment

System	Expected Address	Account	Password
PLC VM	192.168.1.10	plc	password

Note: If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

Procedure

Step 1. Import the ICDFA-provided PLC VM into the virtualization platform specified by your instructor. Configure the laboratory network as Internal/Host-Only unless your instructor provides a different isolated configuration. Do not use Bridged networking.

Step 2. Create a clean snapshot named LAB01-START before making changes. Start the PLC VM and sign in with the assigned laboratory account.

Step 3. Confirm that the VM is using the expected training address 192.168.1.10. If your instructor supplied a revised image or address, record the value in your report instead of changing it without approval.

Step 4. Open the OpenPLC Editor from the desktop/application menu.

Step 5. Select File > Open and browse to the provided PLC Programming Language Examples directory. Open the Ladder Diagram project.

Step 6. Under the Project tree, open Res0 and edit instance0 (Motor_Control). Expand the ladder diagram so the full motor-control logic is visible.

Step 7. Identify the two power rails, the rung(s), the Start contact, Stop contact, MotorOutput contact used for latching, and the MotorOutput coil.

Step 8. Trace the logic from the left rail to the right rail. Explain what must be true for the motor output to energize and what causes it to de-energize.

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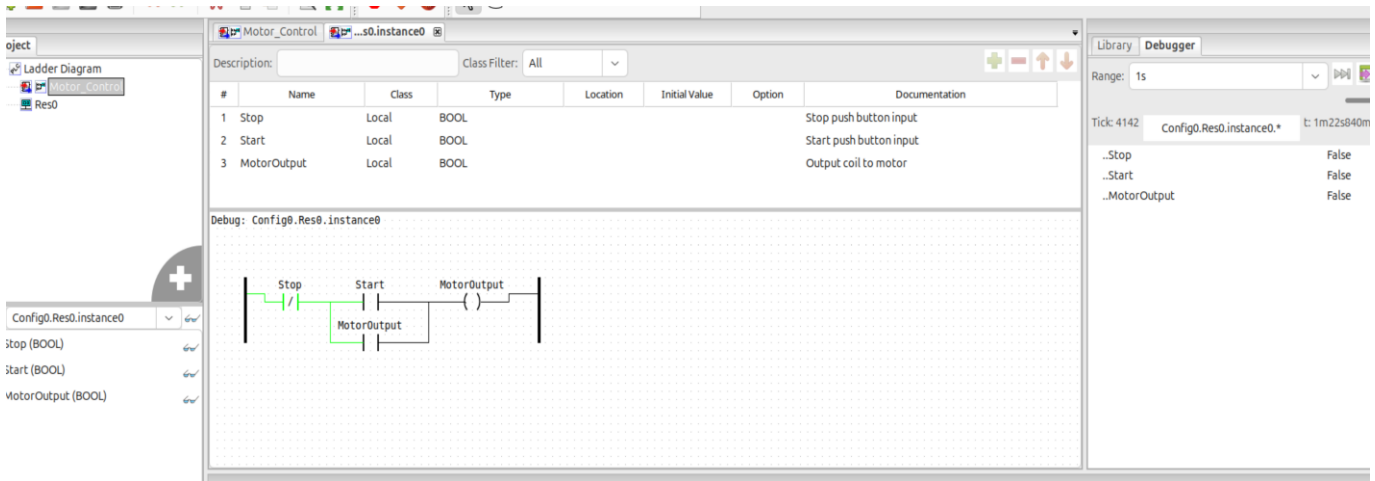


Fig 1: NO Start || is false and NC Stop ||/ is false, hence the motor is de-energized

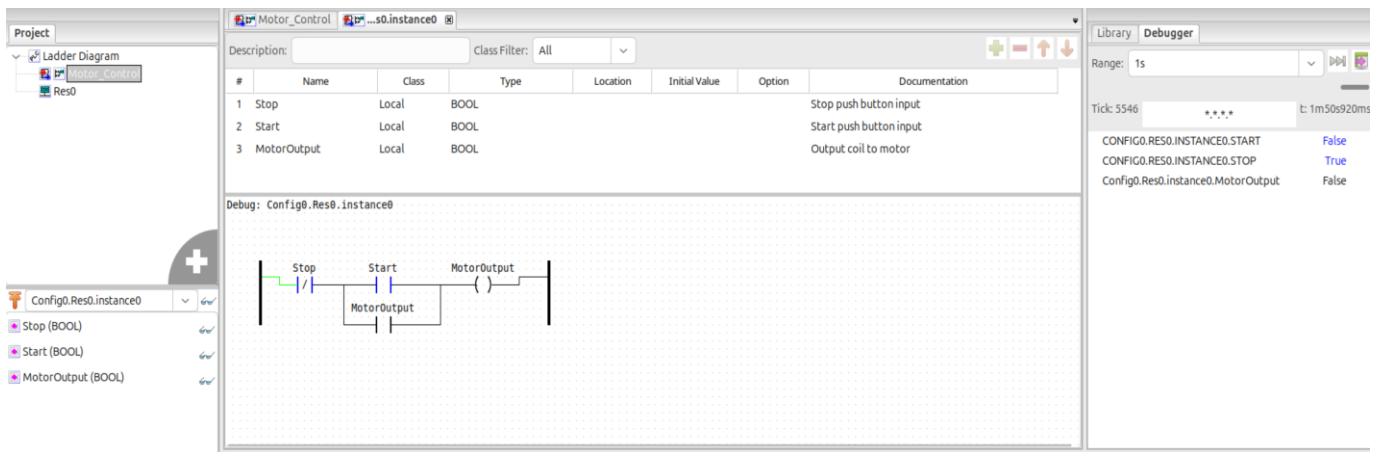


Fig 2: NO Start || is false and NC Stop ||/ is true, hence the motor is de-energized

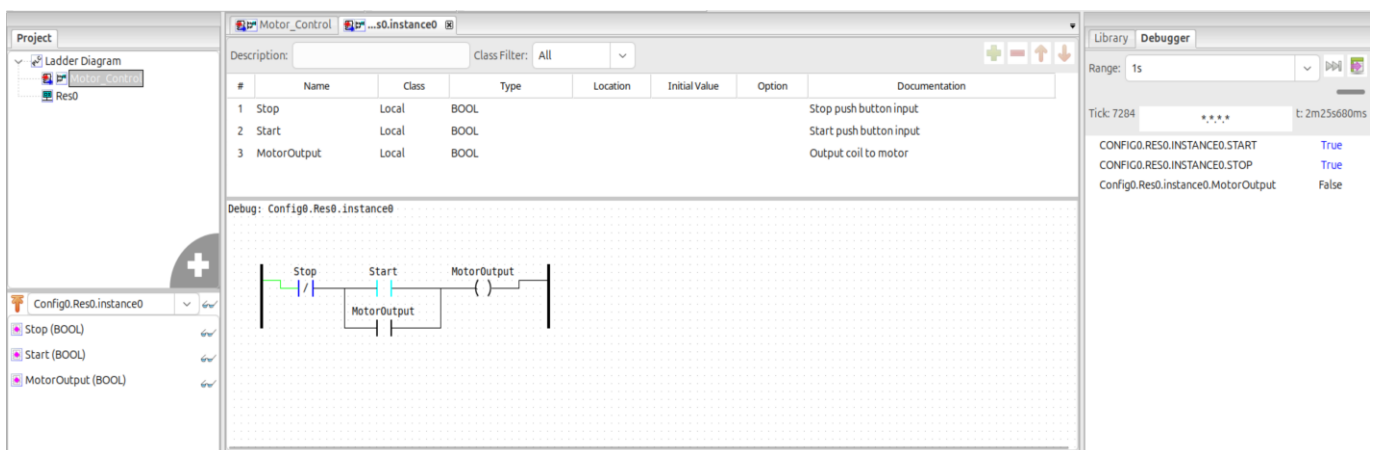


Fig 3: NO Start || is true and NC Stop ||/ is true, hence the motor is de-energized

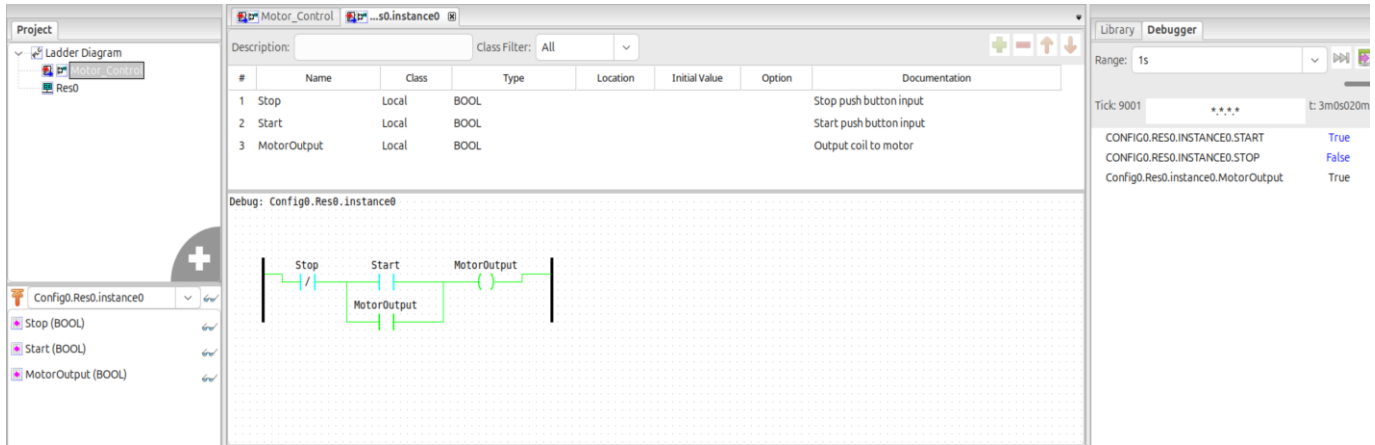


Fig 4: NO Start || is true and NC Stop ||| is false, hence the motor is energized

Step 9. Record one example each of a discrete input, analog input, discrete output and analog output that could appear in an industrial process.

Type	Value
Discrete Input	Toggling a light switch ON or OFF
Analog Input	Temperature readings
Discrete Output	lightbulb is turned ON or turned OFF
Analog Output	A valve is 50% open

Step 10. Capture the required evidence, save your report, then restore or power off the VM as directed by the instructor.

Required Evidence & Submission

- Screenshot showing the OpenPLC Motor_Control ladder diagram.
- A labelled explanation of the Start, Stop, latch and MotorOutput elements.
- A completed I/O classification table with at least four industrial examples.
- A 150–250 word reflection on why network-connected PLCs create both operational benefits and cybersecurity considerations.

Reflection: Why network-connected PLCs create both operational benefits and cybersecurity considerations.

The operational benefits of network-connected PLCs are significant. The core benefits include remote monitoring and diagnostics which allow engineers to view real-time status, alarms and diagnostic data from anywhere which reduces downtime by enabling faster issue identification and resolution. It also allows centralized data collection and analytics which supports trend analysis and predictive maintenance and is essential for compliance and audit reports.

Network-connected PLCs make it easier for firmware to be updated remotely without the need to shut down the site or having the physical presence of an Engineer. It also allows real-time synchronization between PLCs and facilitates job scheduling and makes adding new machines to the existing network simpler.

On the flip side, there are a couple of security considerations that need to be considered. Network connected PLCs introduce cyber threats like malware, ransomware and could lead to a breach which could affect not only sensitive data but the physical operations of the facility. Many PLCs utilize legacy protocols like Modbus TCP which requires no authentication or encryption meaning anyone on the network can read or write to the PLCs if they understand the protocol structure.

Proper access control systems can track who modifies logic or parameters in real time, maintaining audit trails. However, if network segmentation is lacking, there is a risk of lateral movement from a compromised workstation to any of the PLCs

As PLCs are built to last a very long time, firmware may remain unpatched or older models no longer receive security updates from vendors, introducing the risk of a potential exploit.

Recommended filename: C11/26/ACIS_17347_LAB01_Introduction_to_Programmable_Logic_Controllers.pdf

Knowledge Check

1. What is the difference between an input module and an output module?

Response: _An input module receives a signal while an output module sends a signal_

2. Why is a normally closed contact commonly associated with stop or emergency-stop logic?

Response: _Because the machine will keep running until the emergency/stop button is pressed, thereby breaking the circuit and interrupting the flow of current to stop the machine_

3. Give one example of a discrete I/O signal and one example of an analog I/O signal.

Response: _Discrete I/O signal – Push button this is ON when pressed and OFF when not pressed while Analog I/O signal I – A 4-20mA signal to a control valve where the current value determines if the valve should be fully open or fully closed_

4. What operational risk can arise when industrial automation devices become network-connected?

Response: _There is a risk of compromise or unauthorized actions performed on the devices if there is a network breach

Marking Rubric — 100 Marks

Criterion	Marks	Standard
Technical execution	50	Lab procedure completed correctly; required configuration/analysis works as intended.
Evidence	20	Screenshots/files are complete, readable, relevant and clearly labelled.
Analysis & knowledge check	20	Responses demonstrate understanding of the industrial-control concepts observed.
Professional reporting	10	Clear structure, correct terminology, concise writing, file naming and submission compliance.

Submission Integrity: Evidence must come from the student's own assigned laboratory environment. Screenshots should show enough context to verify the activity. Do not submit another learner's screenshots or project files.